

f"hi {name}.title() "

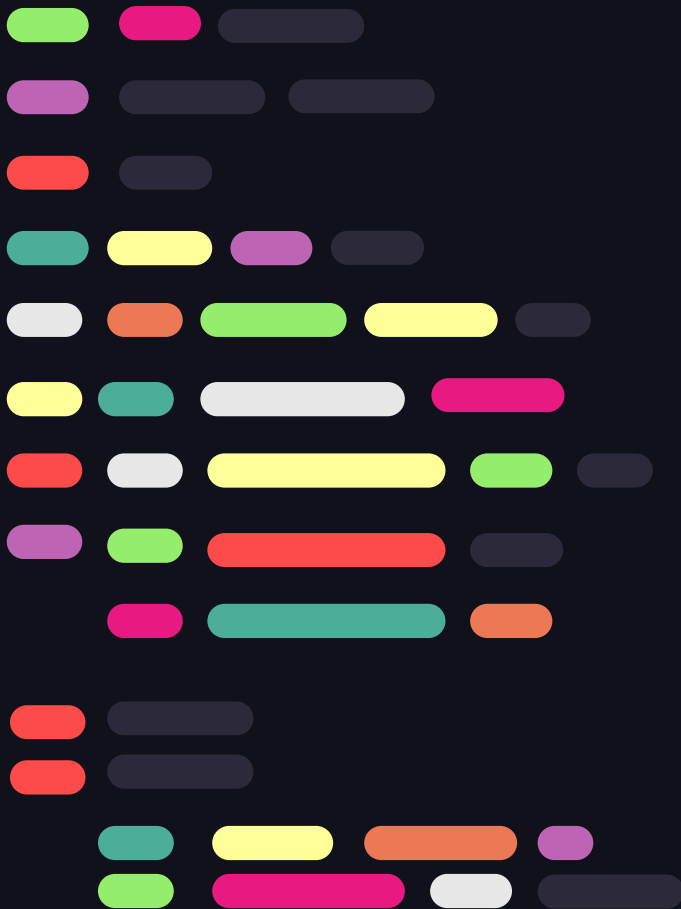


Python Essentials for PCEP

instructor: Alejandro Miranda

continue



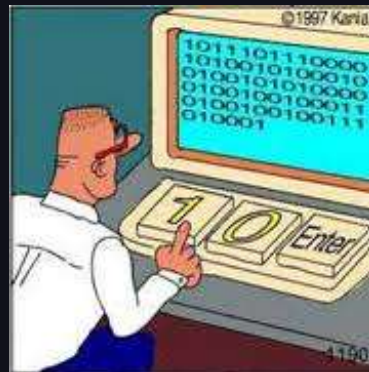


Language

#



What make a language?



Machine
language

- **Alphabet**

a set of symbols used to build words of a certain language

- **Lexis**

a set of symbols used to build words of a certain language

- **Syntax**

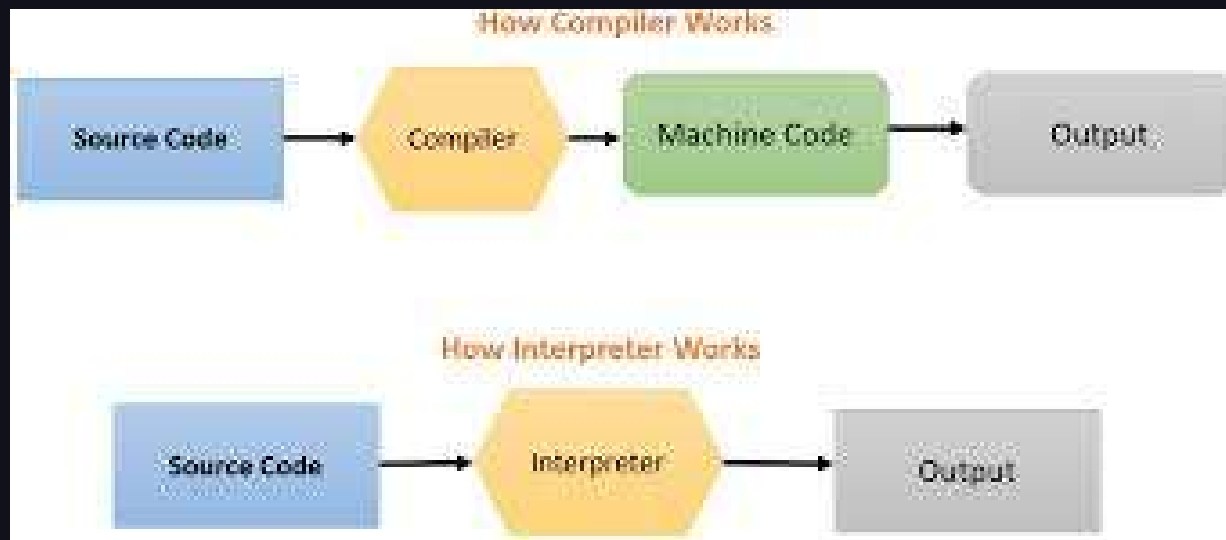
a set of rules used to determine if a certain string of words forms a valid sentence

- **Semantics**

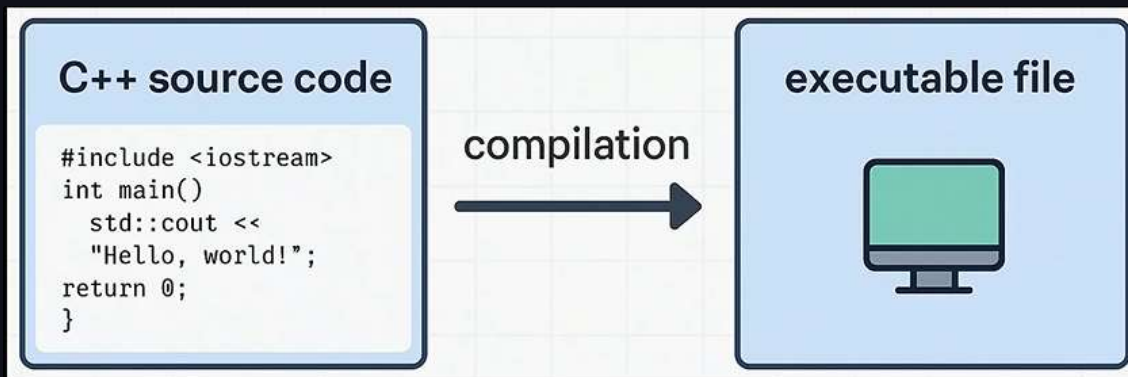
a set of rules determining if a certain phrase makes sense



Compilation vs Interpretation

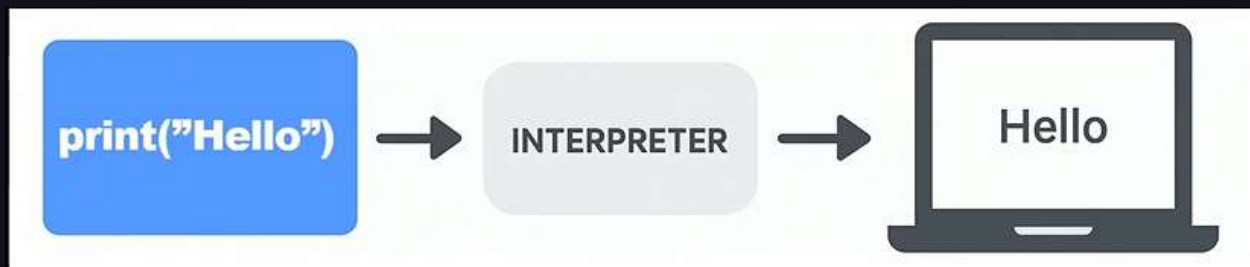


Compilation

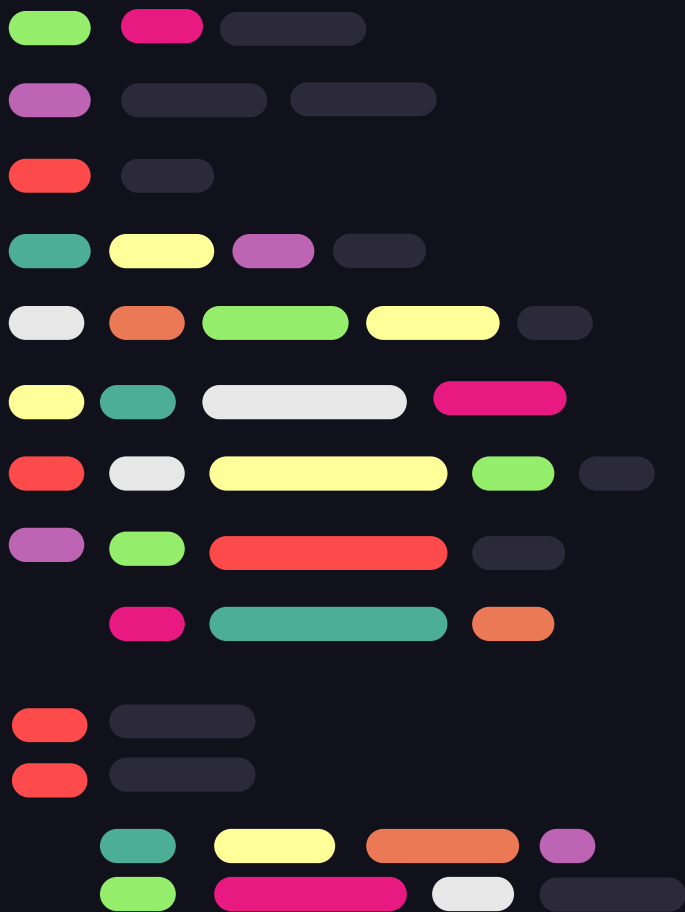


advantages	Disadvantages
Execution usually faster	Compilation time
End user needs EXE only	Amendments
Stored as machine language	Multiple hardware platforms

Interpretation



advantages	Disadvantages
Run immediately	Speed
Multiple Hardware platforms	Stored as high-level language
	Everyone needs the interpreter





- Guido Van Rossum
- 1992
- Monty Python
- Characteristics
 - High level
 - Dynamic typing
 - OO
 - Interpreted
 - Open source
 - Multi platform





Key Application Areas of Python

Artificial Intelligence and Machine Learning:

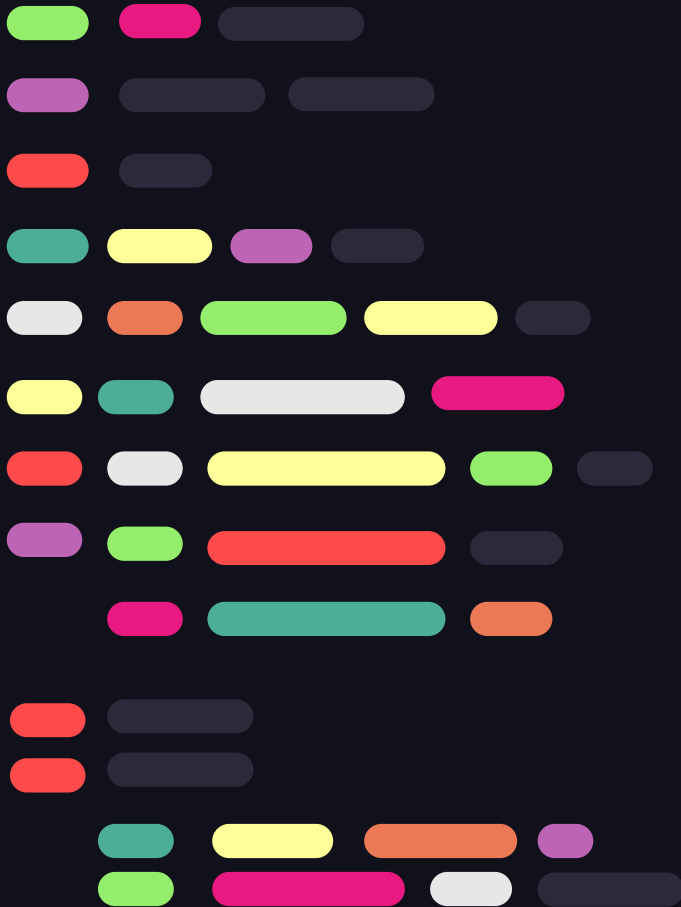
Building smart models and neural networks using libraries like PyTorch, TensorFlow, and scikit-learn.

Data Analysis and Science: Cleaning, processing, and viewing large amounts of data with tools like Pandas, NumPy, and Matplotlib.

Web Development: Creating server-side backend logic and APIs for websites using frameworks like Django and Flask.

Automation and Scripting: Writing small programs to handle repetitive system administration tasks and file management





Python Installation

#





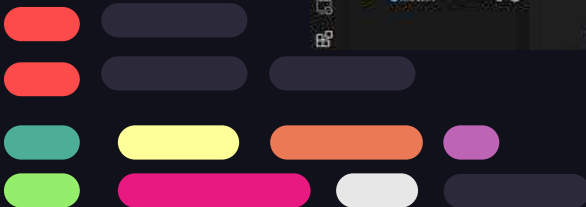
- **interpreted dowload**

<https://www.python.org/>

- **VSC (Visual Studio Code) dowload**

https://code.visualstudio.com/download?_exp_download=d53503e735

- **Extention Python for VSC**





First steps with python

#



My first contact on python

- Variables
- Literals
 - Boolean
 - Integer
 - Floating



Machine
language





- **Keywords**

- **Instructions**

```
X = 10  
print("Hello world! ")
```

- **Indentation**

- **comments**



Keywords

and	as	assert	break	class
continue	def	del	elif	else
except	for	from	global	if
import	in	is	lambda	not
or	pass	raise	return	try
while	with	yield		

